

ABHIJIT MANDAL

Bengaluru, India | +91-7352690161 | abhijit.mandal777@gmail.com | linkedin.com/in/mandalabhijit | github.com/Abhijit-Mandal

SUMMARY

Backend Engineer with nearly 3 years of experience designing and scaling distributed SaaS systems handling multi-tenant architectures across global regions. Strong ownership of microservices, event-driven systems, and production reliability, with hands-on experience building high-scale workflow engines, tenant lifecycle systems, and real-time data pipelines using Java and Spring WebFlux.

TECHNICAL SKILLS

Languages: Java, SQL, Groovy, Bash

Frameworks: Spring Boot, Spring WebFlux, jOOQ, Flyway, Spock (Groovy)

Databases: MySQL, PostgreSQL, Redis

Cloud & Infrastructure: AWS (SQS, S3, EventBridge), Docker, Kubernetes, Jenkins

Observability: OpenTelemetry, Micrometer, Grafana

Tools: Git, Gradle, Linux

Concepts: REST APIs, Microservices, Event-driven systems

WORK EXPERIENCE

LeadSquared

Aug 2023 – Present

Software Engineer (Backend)

Bengaluru, India

- Built **Tenant Management Services** for **Service Cloud** and **Customer360** platforms, handling tenant provisioning, configuration, feature lifecycle, and billing management for **300+ enterprise tenants** across **SGP, US, and India** regions using **Java, Spring WebFlux, and Redis**.
- Designed and owned **Process Builder microservice** from scratch, building a workflow automation engine handling **100K+ executions/month**, using **Redis Streams** for async event ingestion and **AWS EventBridge + SQS** for SLA scheduling and stage escalation workflows.
- Implemented Kafka-based event pipelines integrating with internal analytics system **Siera**, enabling real-time ingestion of **25K-50K events/day** across **50+ tenants**, and developed **audit and reporting system**, powering **system-defined and custom reporting** that helped customers track **team, agent, and organization-level performance** across **200+ enterprise accounts**.
- Designed a **feature pricing and plan management system** enabling tenant-level feature control, usage tracking, and **downgrade orchestration** across **20+ dependent services**, implemented phased rollout and backward-compatible schema changes to avoid downtime.
- Enabled **Microsoft OAuth2 authentication** integrated with **JWT-based authorization** for tenant metrics and data migration applications, enabling **role-based access control (RBAC)** across organization-wide systems and securing access to critical operational workflows.
- Implemented **tenant lifecycle cleanup system** in TMS to automatically purge stale tenant data (OpenSearch, configuration DBs, and tenant-specific databases) after inactivity thresholds, using **AWS EventBridge + SQS** for scheduled cleanup workflows with safeguards to prevent deletion of reactivated tenants, reducing infrastructure costs.
- Owned **production lifecycle and reliability** across multiple backend services (TMS, Process Builder, BFF), managing **schema evolution** using Flyway and jOOQ, **observability, alerting, and debugging** using Grafana, and **end-to-end release management** for features and production fixes.

LeadSquared

Mar 2023 – Aug 2023

Software Engineer Intern – Backend

Bengaluru, India

- Contributed to early development of **Tenant Management Service**, implementing tenant lifecycle flows, callback APIs, and async workflows with error handling and duplicate request detection
- Designed and shipped **Ops Dashboard APIs** for tenant listing, status monitoring, and configuration management, powering internal dashboards built using **Budibase** for real-time tenant health visibility

PROJECTS

AI Coach – Fitness Application

Java, Spring Boot, Spring Security, JPA, MySQL, Redis, OpenRouter (Mistral 7B)

- Built a **stateful workout tracking system** with session-based lifecycle management, enabling reuse of historical exercise data for progress tracking.
- Designed APIs for **data persistence and pre-population**, improving tracking consistency and reducing manual input across sessions.
- Integrated an **LLM-based assistant** (Mistral 7B via OpenRouter) and optimized performance using **Redis caching**.
- Implemented **authentication and RBAC** using Spring Security with efficient relational data modeling via JPA.

EDUCATION

Institute of Technical Education and Research

Bachelor of Technology in Computer Science and Engineering

Bhubaneswar, India

2019 – 2023